

Laboratory of Translational Oncology
Medical University of Gdańsk
Gdańsk, Poland
abednarz@gumed.edu.pl

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Dr Pooja Jadhav
Assistant Editor
Scientific Reports

Dear Dr Jadhav,

Please find enclosed the revised manuscript entitled “An Accuracy and Calibration Benchmark of scGPT Pipelines and Matched Classical Models for Cell Type Annotation” (submission ID 12ef9a8a-5497-4180-b23f-174604d3eec4). The title contains 16 words and no punctuation, in accordance with the editorial request.

We have addressed the reports through a new donor-aware statistical design and a complete standalone rewrite. The primary analysis now uses five donor-held-out folds in three human atlases, covering 965,427 analysed cells and 312 donor identifiers. Frozen official scGPT 0.2.5 embeddings and normalized expression of the same training-selected genes are each paired with Logistic Regression, Random Forest and XGBoost. Hyperparameters are selected separately inside the training donors, and primary inference compares representations at a fixed classifier. SVD-512 provides a secondary dimensionality control for every atlas with Logistic Regression and for TNBC and brain with all three classifiers. We additionally provide donor-cluster bootstrap intervals, class-specific error analyses, labelled-only feature selection under scarcity, and held-out-donor temperature scaling for Logistic Regression and XGBoost in TNBC and brain. The human-checkpoint pig analysis is separated as a cross-species stress test and is not included in the primary claim.

The manuscript consistently distinguishes a frozen scGPT representation from the downstream classifier that produces annotation probabilities. Its conclusions are limited to the evaluated pipelines, portal labels and within-atlas unseen-donor setting; they do not extend to official fine-tuning, other checkpoints, other foundation models, zero-shot discovery or external-atlas transfer.

The submission includes a clean manuscript, a marked manuscript, Supplementary Information, a detailed point-by-point response, separate publication-quality figures, complete LaTeX source, compact numerical results and executable analysis code. Dataset identifiers, software versions, source commit, checkpoint hashes and exact five-fold pipeline coverage are recorded for reproducibility.

Thank you for considering this revision. We believe the donor-held-out factorial design and expanded calibration analysis directly address the methodological concerns and substantially strengthen the paper.

Sincerely,

Anna Supernat
on behalf of all authors